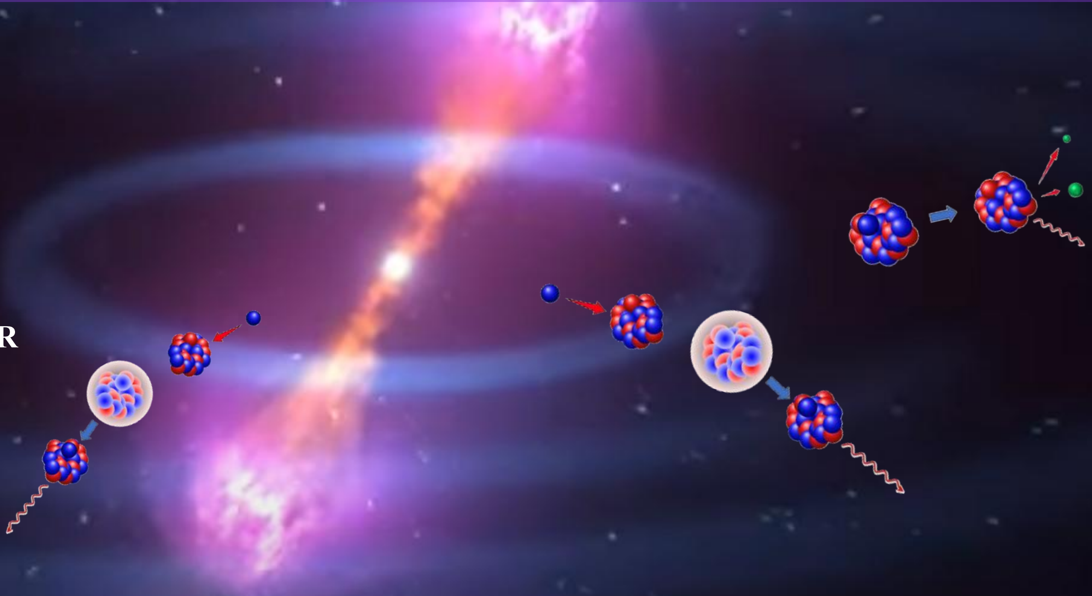
Correlated Monte Carlo of neutron capture rates in the weak *r* process

ATUL KEDIA
POSTDOC

APS MEETING, DENVER
MARCH, 2026



LDRD

NC STATE UNIVERSITY

Kedia+2026 arXiv:2602.12428

NS merger illustration: NASA Goddard

Collaboration

Lawrence Livermore National Lab

- Jutta Escher
- Kostas Kravvaris
- Jeff Berryman
- Cole Pruitt
- Erika Holmbeck
- Andre Sieverding
- Oliver Gorton

e^-

(n,γ)

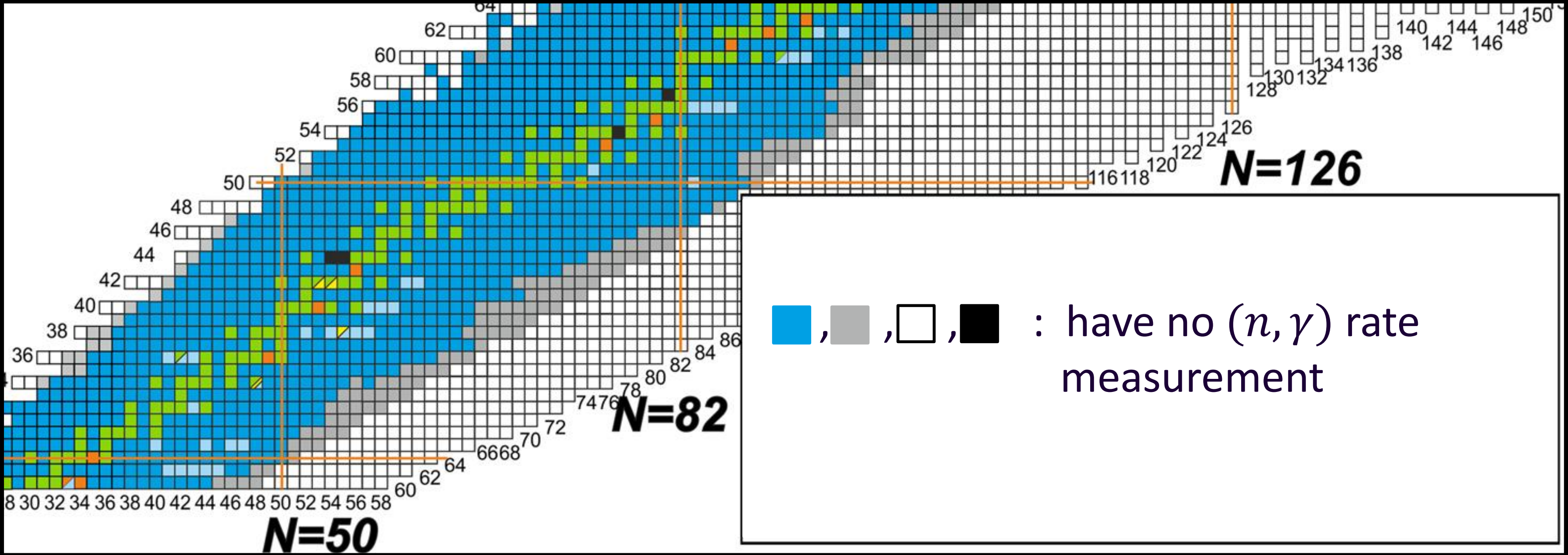
University of Notre Dame

- Rebecca Surman
- Jonathan Cabrera Garcia

North Carolina State University

- Gail McLaughlin
- Yanlong Lin
- Paul Toolan
- Atul Kedia

State of Neutron capture rate measurements



Adapted from Dillman et *al.*, *Eur. Phys. J. A* (2023)

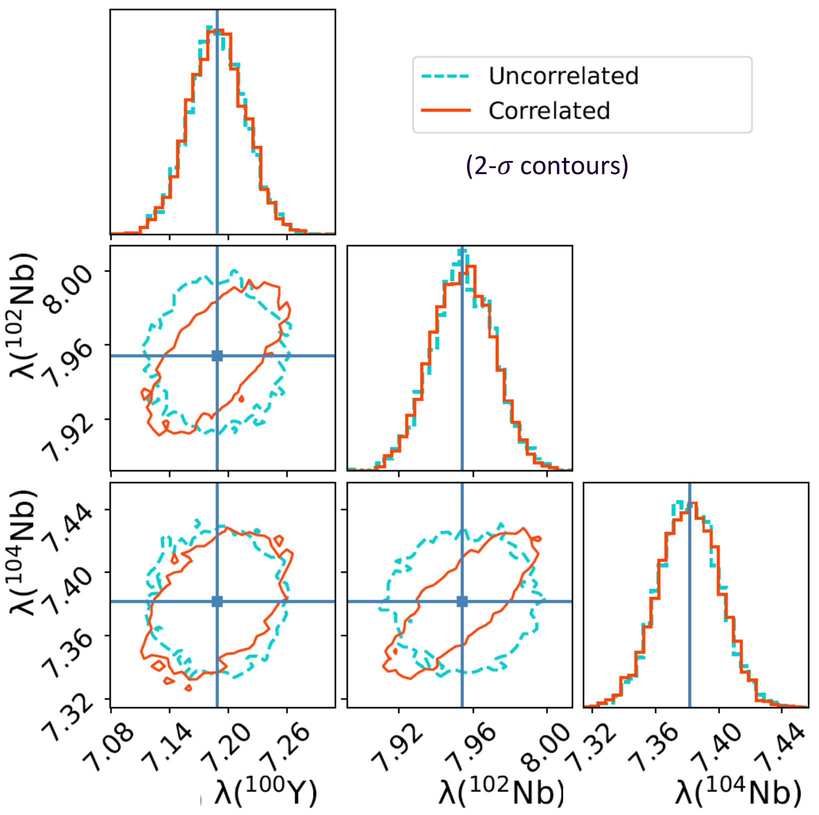
Uncorrelated v Correlated Monte Carlo variations

Uncorrelated samples :

Neutron capture rates of each isotope is sampled independently of others.

Correlated samples:

Neutron capture rates samples are correlated due to underlying nuclear theory inputs, the optical model potential: uncertainty quantified Koning-Delaroche (KD), Pruitt et *al.*, Phys. Rev. C (2023).



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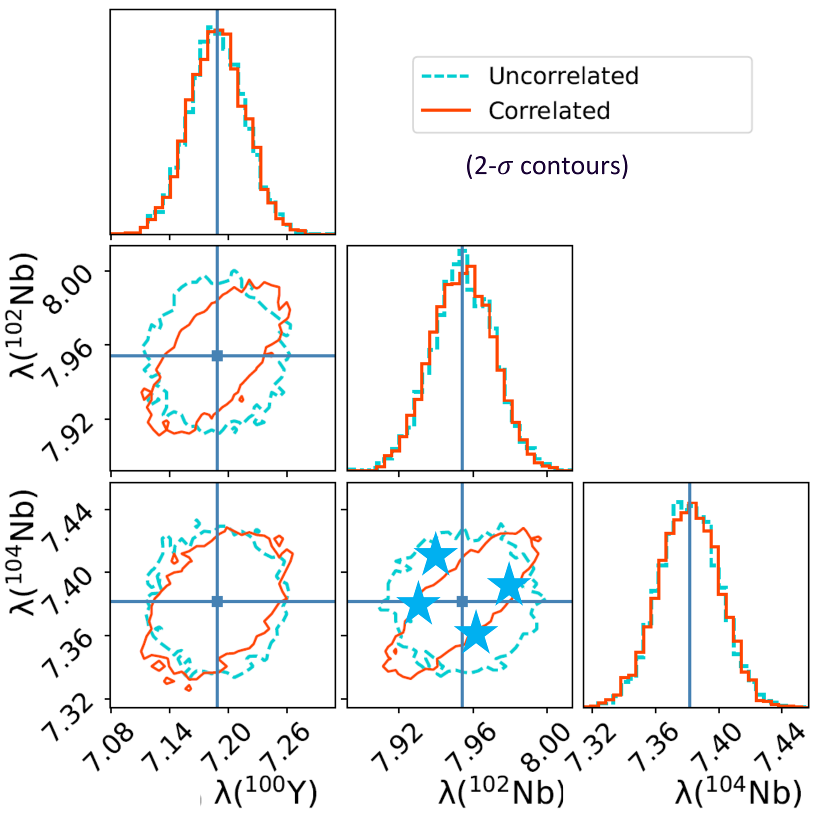

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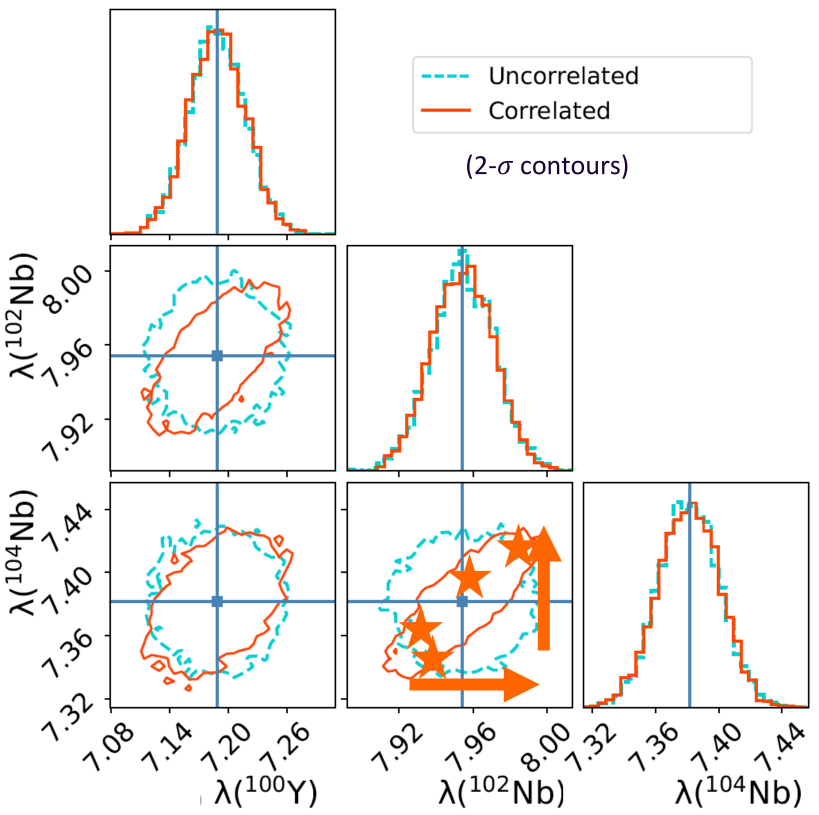
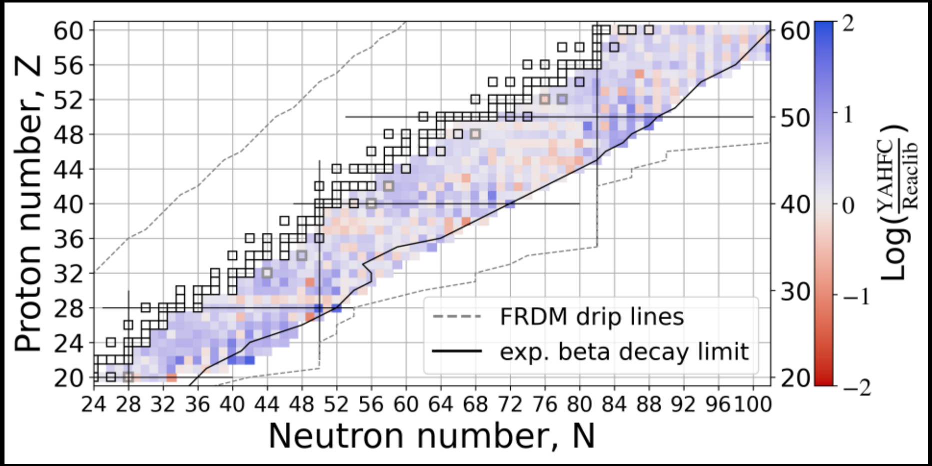


Nuclear and Astrophysics inputs

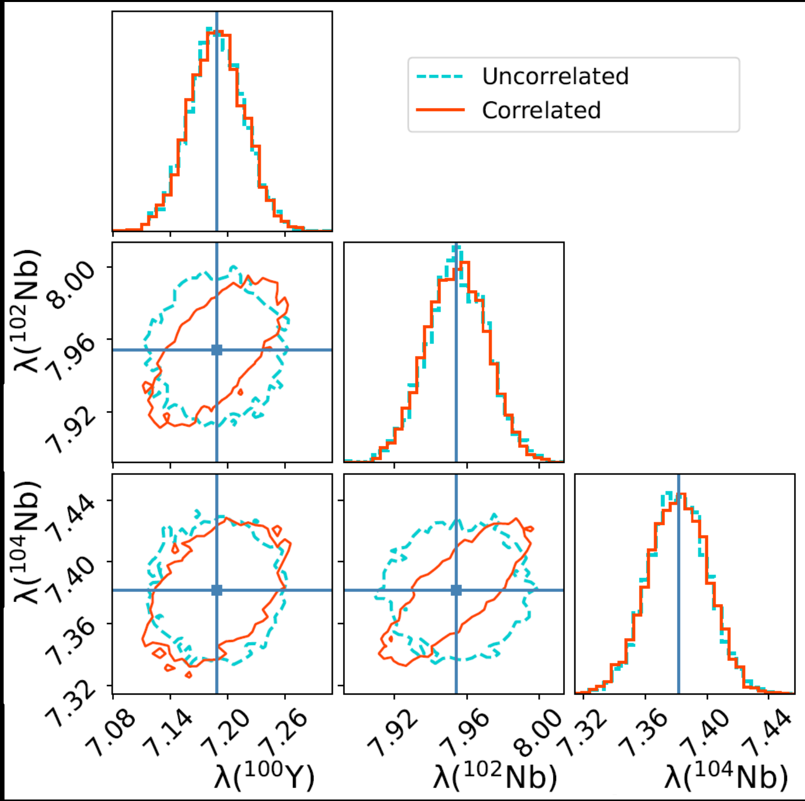
Neutron capture reaction rates
YAHFC - E. Ormand (LLNL) (Yet Another Hauser-Feshbach Code)
1000 isotopes in Z = 20 to 60
Correlated Monte Carlo variations of neutron capture rates

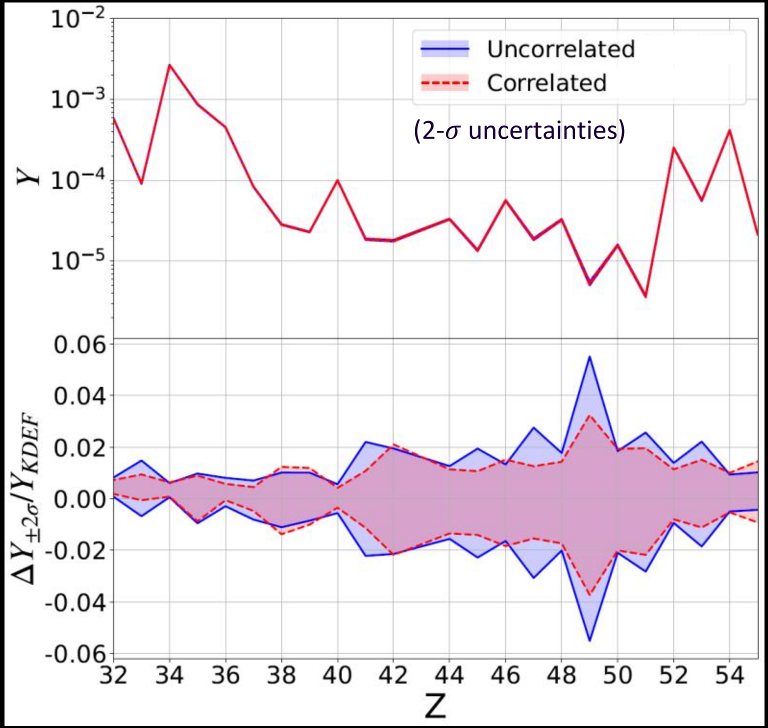
Astrophysical condition
Weak *r* process conditions from a Post merger disk winds simulation (Metzger and Fernandez, MNRAS, 2014)

Reaction network
PRISM
Sprouse, Mumpower, Surman, Phys. Rev. C (2021)



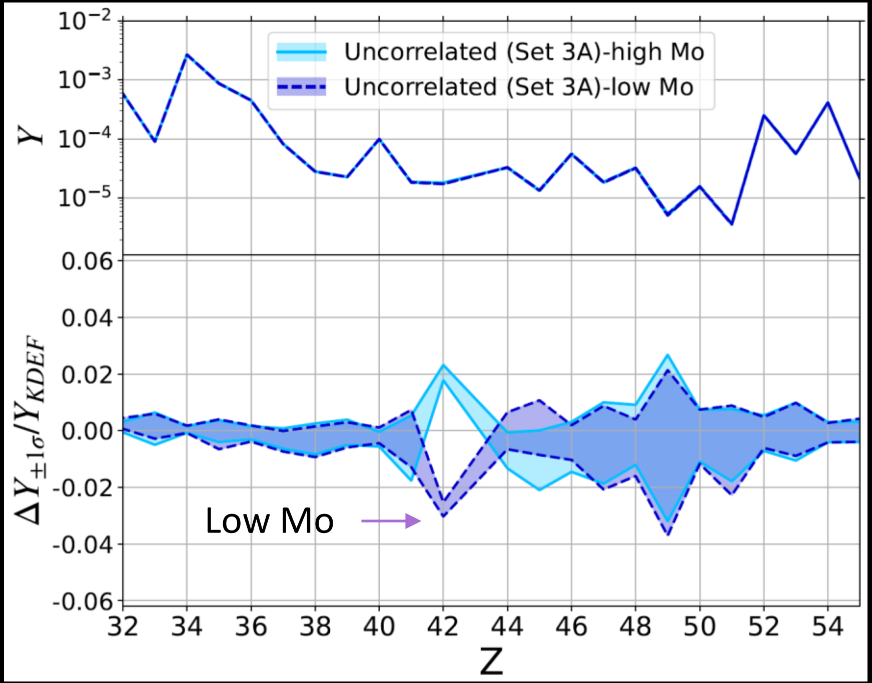
Uncorrelated v Correlated Monte Carlo Uncertainties





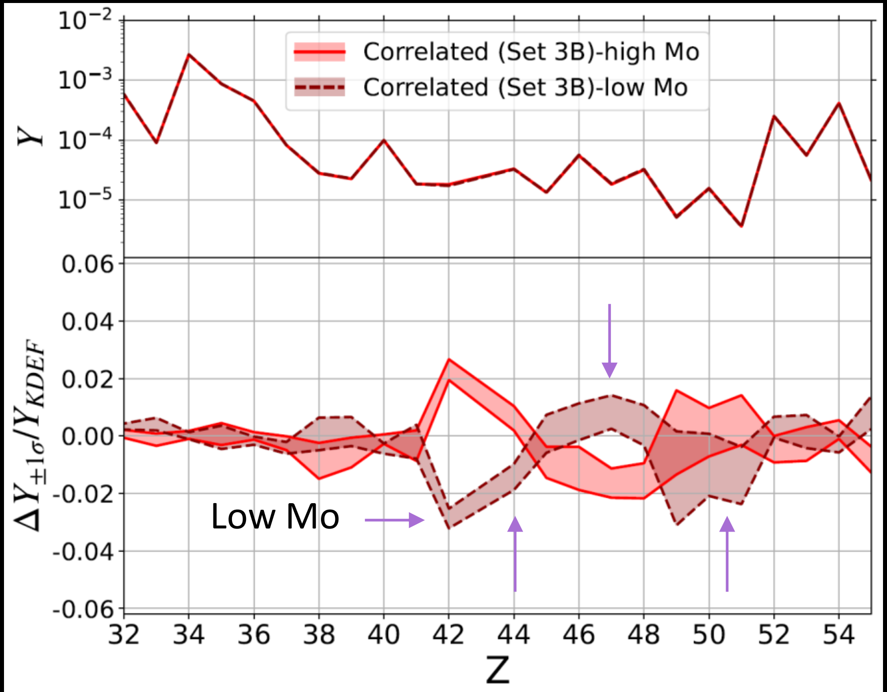
Neutron capture rate correlations due to optical model potential **do not** reduce uncertainties significantly.

Uncorrelated v Correlated: Impact to elements



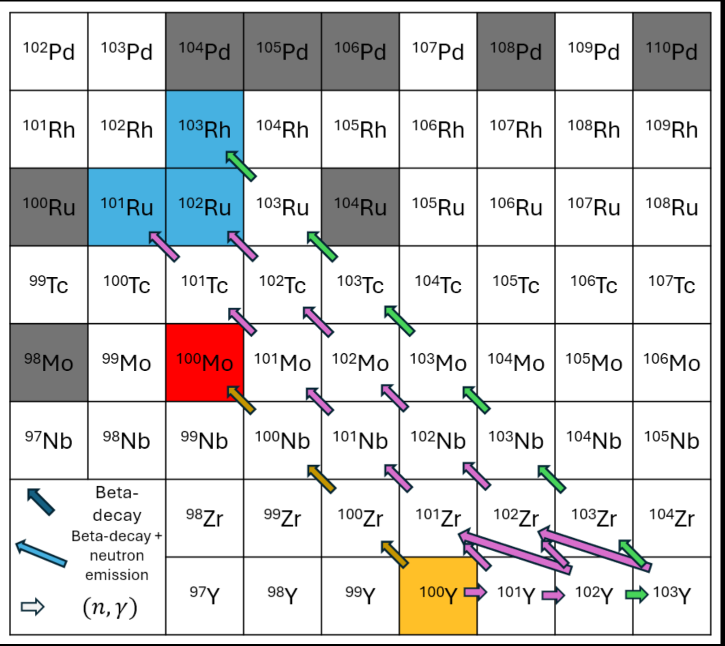
Uncorrelated: Simulations with low Mo (42) increase the abundances of Ru(44) and Rh(45) slightly.

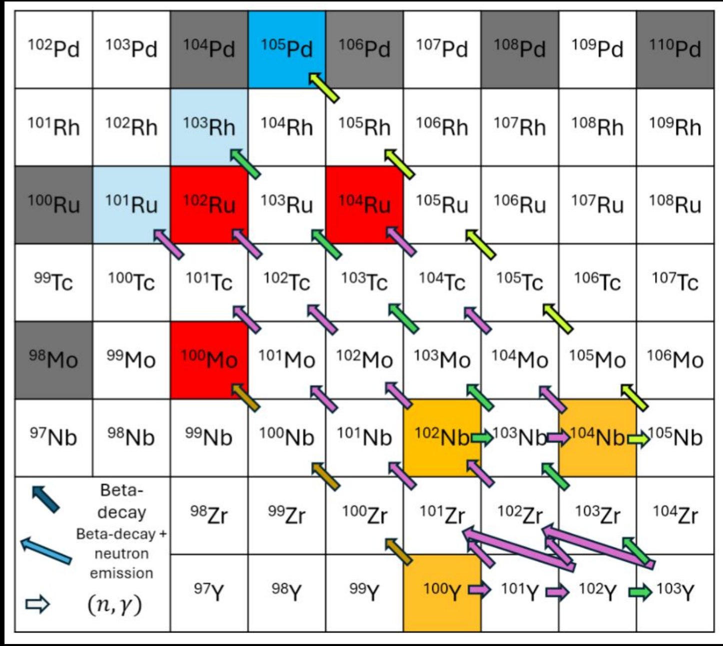
Kedia, et *al.*, arXiv:2602.12428 (2026)



Correlated: Simulations with low Mo (42) also produce low Ru (44), Sn(50), and Sb(51) and high Rh (45), Pd (46), Ag (47), and Cd (48).

Uncorrelated v Correlated: Flow of abundance





^{100}Mo abundance is strongly dependent on ^{100}Y(n,γ) rate. If ^{100}Y(n,γ) rate increases, ^{100}Mo decreases and ^{101}Ru, ^{102}Ru, and ^{103}Ru increase.

Kedia, et *al.*, arXiv:2602.12428 (2026)

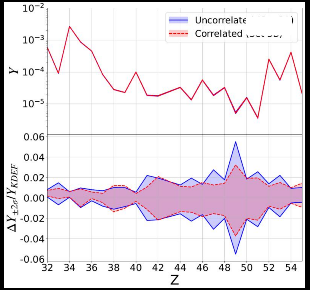
Rate correlations increase ^{100}Y, ^{102}Nb and ^{104}Nb (n,γ) rates together and push material to higher Z elements. Together ^{100}Mo, ^{102}Ru, and ^{104}Ru decrease and ^{101}Ru, ^{103}Rh, and ^{105}Pd increase.

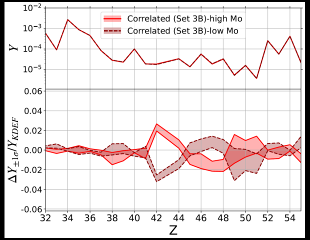
Summary

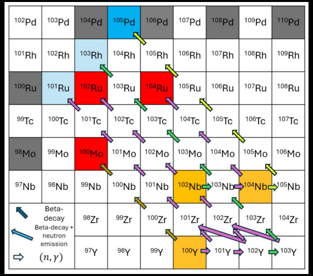
- We've conducted the first correlated Monte Carlo study of neutron capture rate variations in weak *r* process.
  - Correlations associated with optical model potential
- Overall, reaction rate correlations did not reduce uncertainties in weak *r* abundances substantially.
- Element formation flows change substantially due to neutron capture rate correlation.

Future: Correlation in neutron capture rates due to nuclear structure inputs (level density, gamma strength function) still need to be investigated.

AK, Berryman, Cabrera Garcia, Escher, Gorton, Holmbeck, McLaughlin, Pruitt, Sieverding, Surman (arXiv:2602.12428)

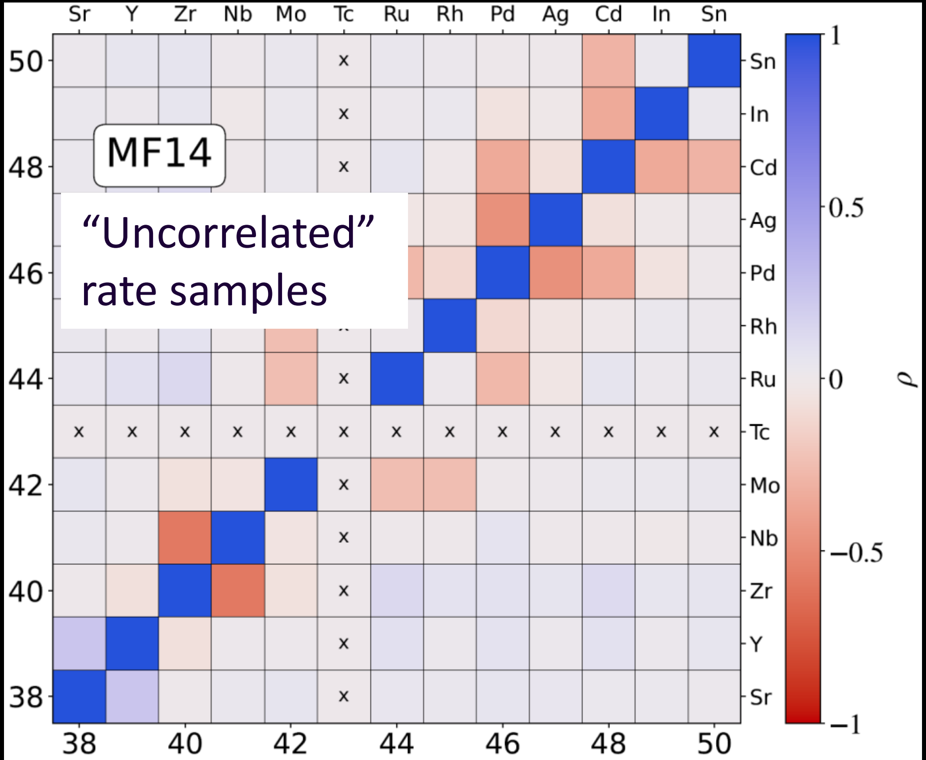


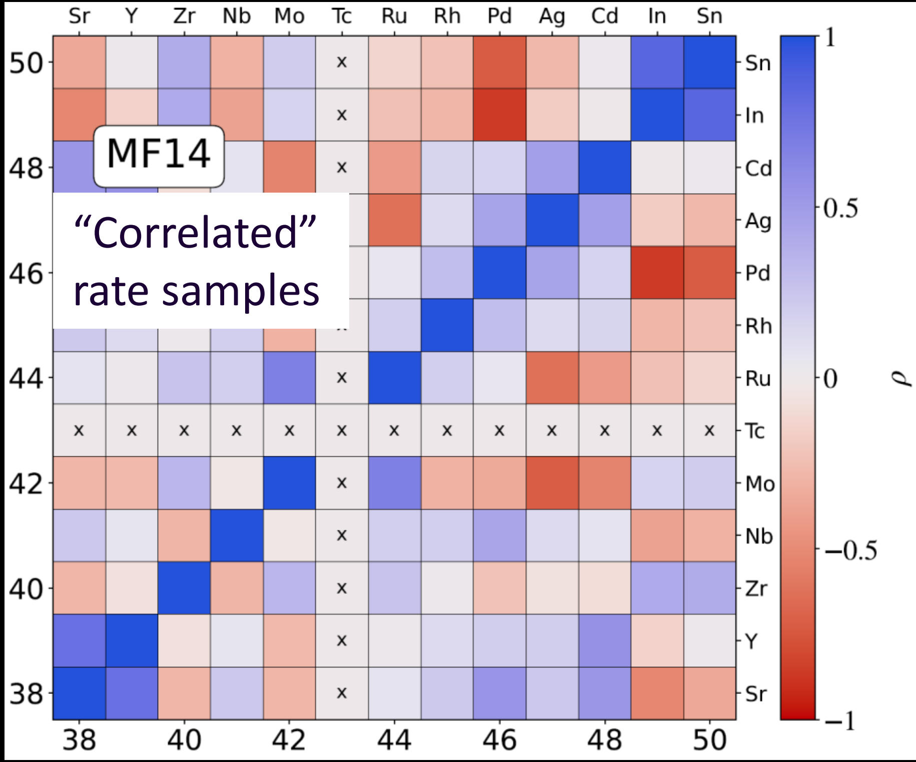




Extra slides

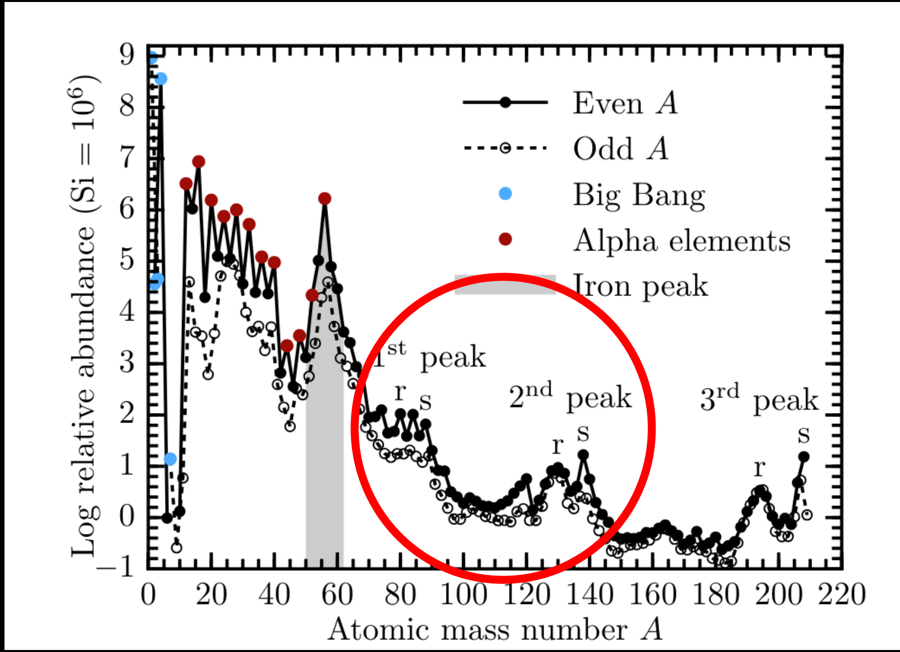
Correlation between abundances



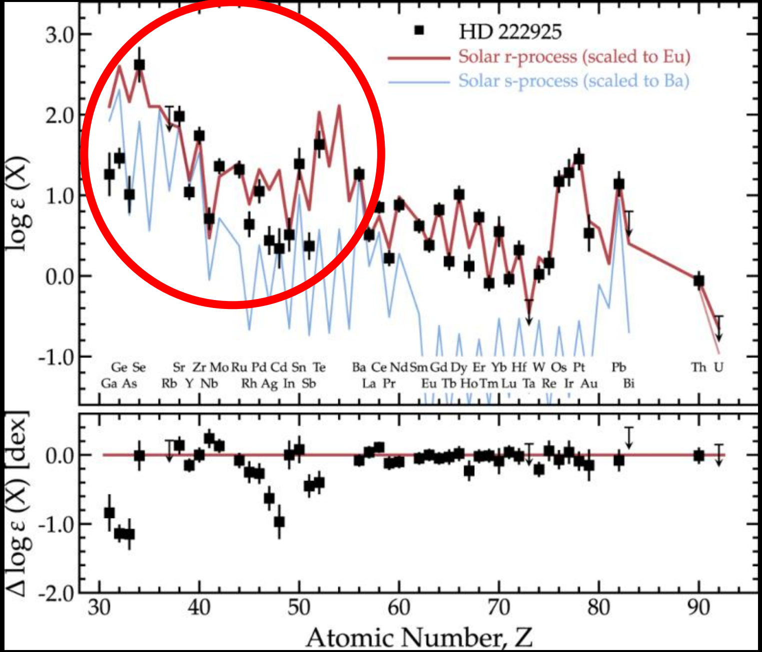


Impactful neutron capture isotopes for the weak r process

Weak r process

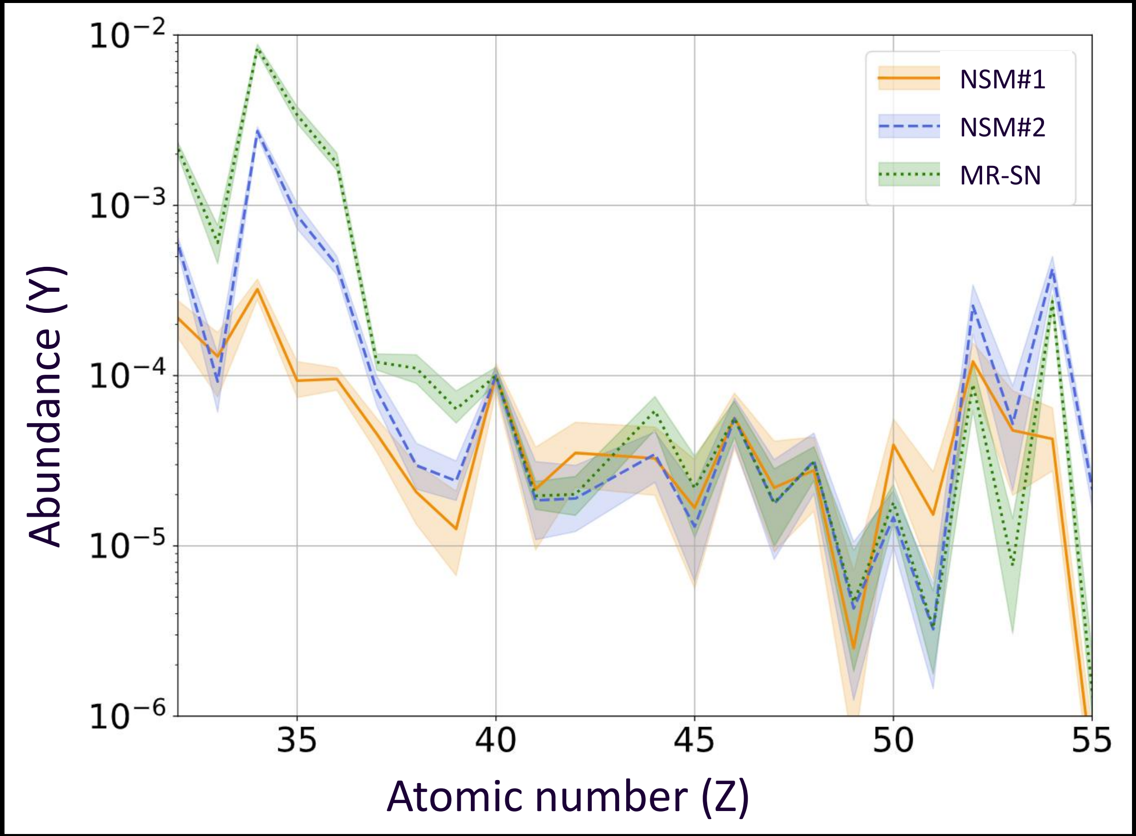


[Lippuner (2018)]



Roederer et *al*, ApJ (2022)

Uncertainties in weak r-process abundances

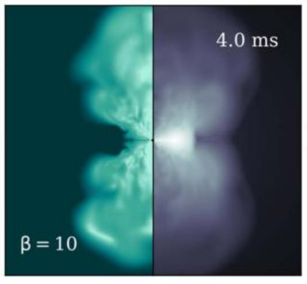


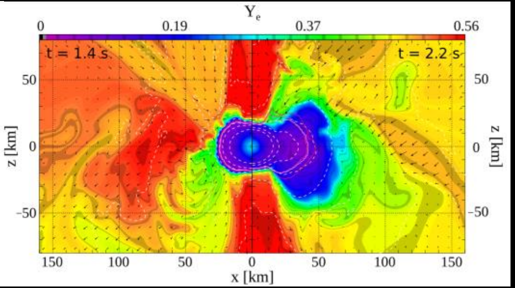
2σ (95 percentile) uncertainty in abundances due to uncertainties in neutron capture reaction rates.

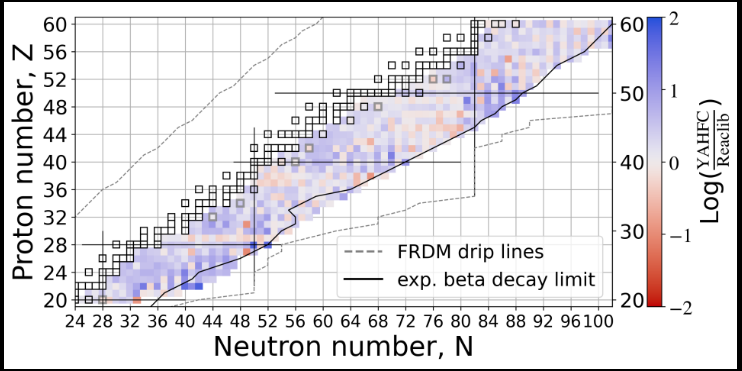
Uncorrelated Astro and Nuclear physics inputs

Astrophysical condition	Simulation
Post Neutron star merger disk wind	K. Lund et al, ApJ, 2024 (νbhlight)
Post Neutron star merger disk winds	Metzger and Fernandez, MNRAS, 2014
Magnetorotional Supernovae	Reichert et al, MNRAS 2021

Neutron capture reaction rates
YAHFC (Yet Another Hauser-Feshbach Code)
Z = 20 to 60 (1000 isotopes)
Monte Carlo variations of neutron capture rates within $O(1)$



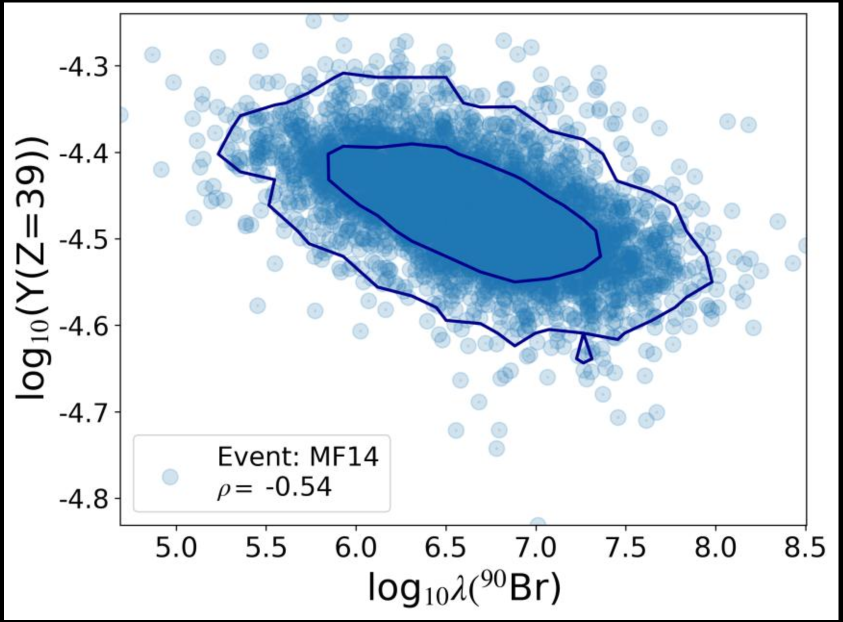


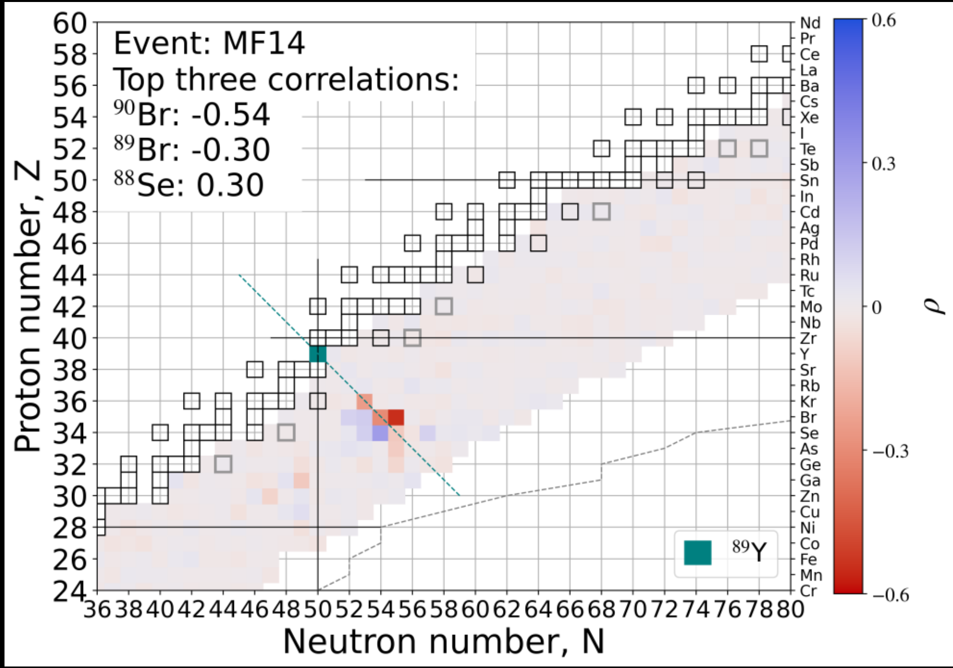


Can the uncertainties in *r*-process elemental abundances due to nuclear reaction rates be reduced?

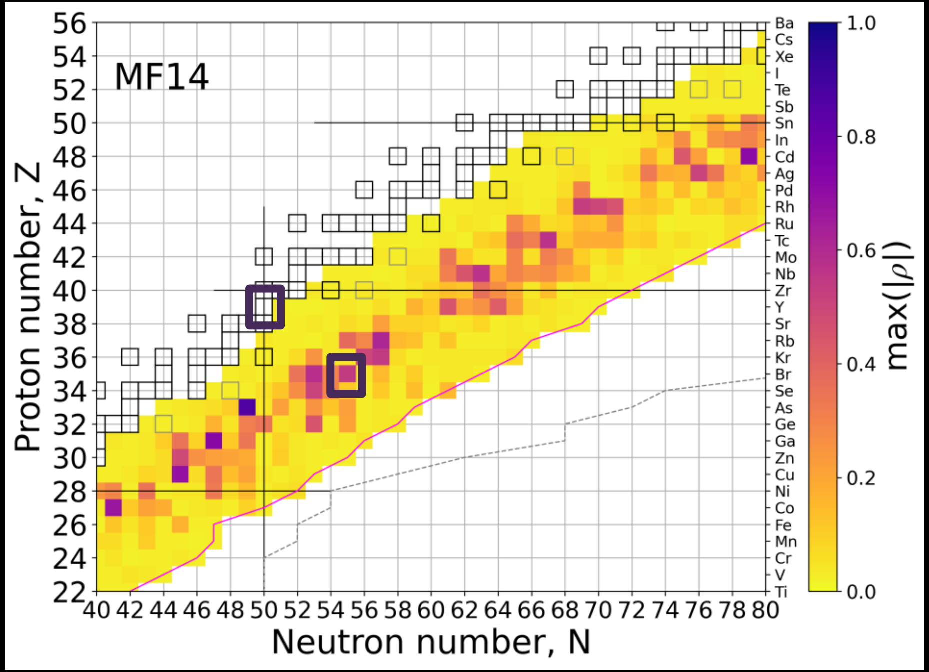
Uncertainties in weak *r*-process element abundance (Y_i) $\propto$ Uncertainties in neutron capture reaction rate (λ)

Element-rate correlation





Most influential neutron capture rates

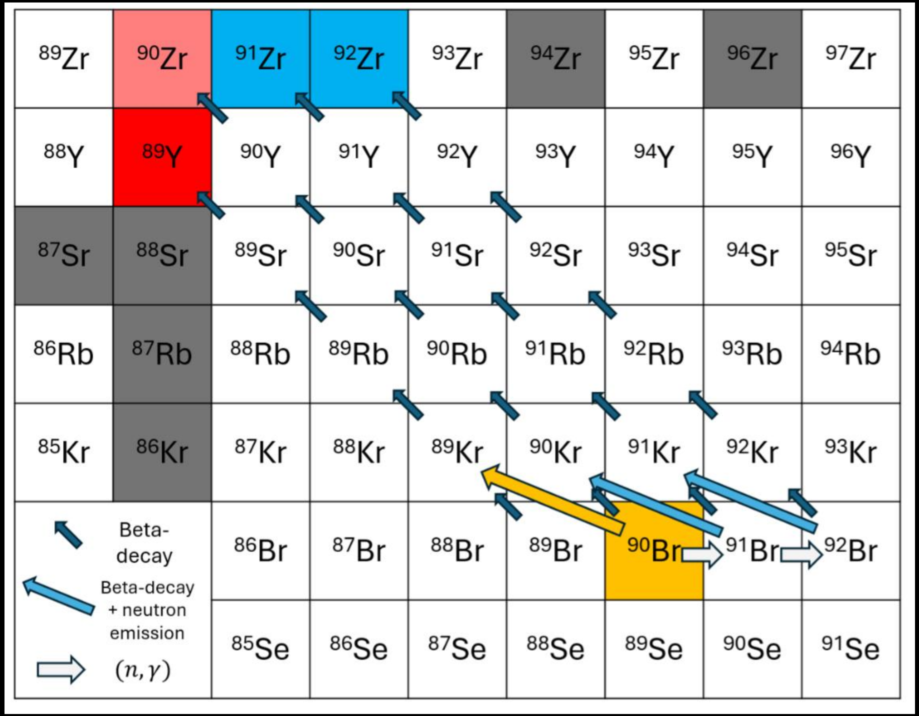


Element	(n,γ) isotope, correlation coeff. (MF14)					
Se	^{78}Ga	-0.65	^{80}Ga	-0.33	-	
Br	^{78}Ga	0.72	^{80}Ga	0.34	^{81}Ge	-0.32
Kr	^{82}As	0.86	^{82}Ge	0.35	-	
Rb	^{87}Se	-0.51	^{85}Ge	-0.45	-	
Sr	^{88}Br	-0.58	^{87}Br	0.35	-	
Y	^{90}Br	-0.54	^{89}Br	-0.30	-	
Zr	^{92}Kr	-0.47	^{94}Rb	-0.40	-	
Nb	^{93}Kr	-0.54	^{92}Kr	0.50	^{92}Rb	0.38
Mo	^{94}Rb	0.61	^{100}Y	-0.45	-	
Ru	^{104}Nb	-0.57	-		-	
Rh	^{103}Nb	-0.45	^{103}Y	-0.42	^{103}Zr	-0.36
Pd	^{110}Tc	-0.47	^{104}Nb	0.34	-	
Ag	^{108}Tc	0.37	^{109}Tc	-0.32	^{109}Mo	-0.32
Cd	^{110}Tc	0.57	-		-	
In	^{114}Rh	0.53	^{115}Rh	-0.50	-	
Sn	^{116}Rh	0.46	^{124}Ag	-0.34	-	
Sb	^{123}Ag	-0.51	^{123}Cd	-0.44	^{120}Ag	0.33
Te	^{130}In	-0.62	^{129}Cd	0.33	-	
I	^{127}Cd	-0.71	-		-	
Xe	^{130}In	0.62	^{129}Cd	-0.38	^{129}Sn	-0.32

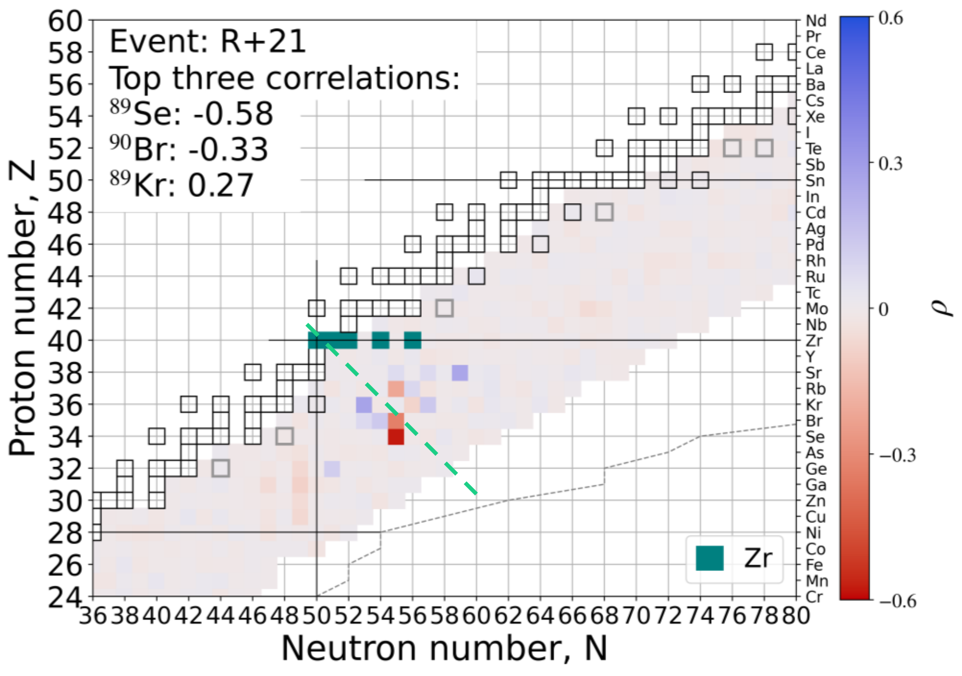
Kedia, et *al*., arXiv:2602.12428 (2026)

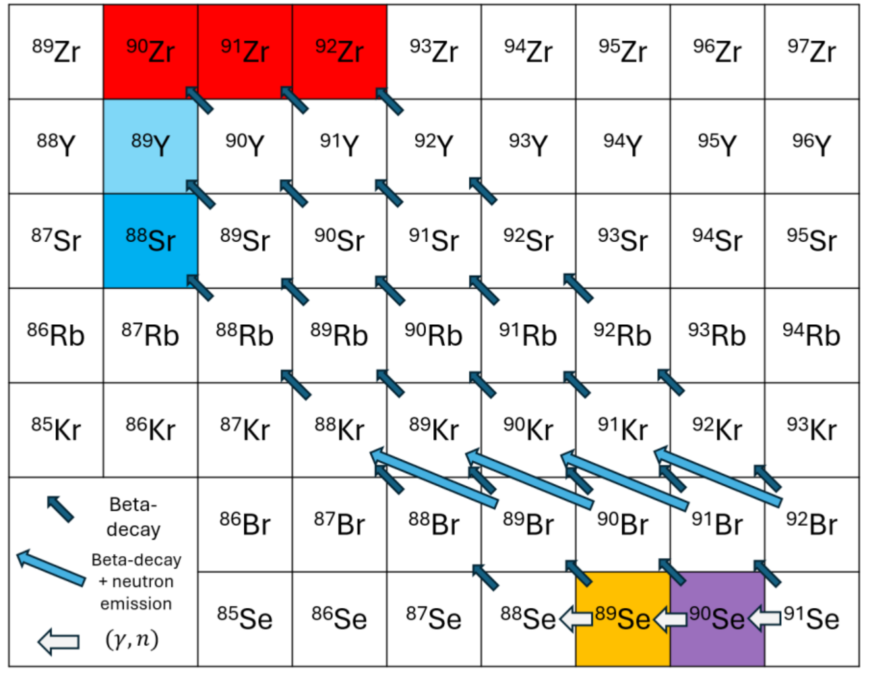
Yttrium – ^{90}Br

^{90}Br has a 25% probability of undergoing a beta-delayed neutron emission and impacts **Yttrium** formation.



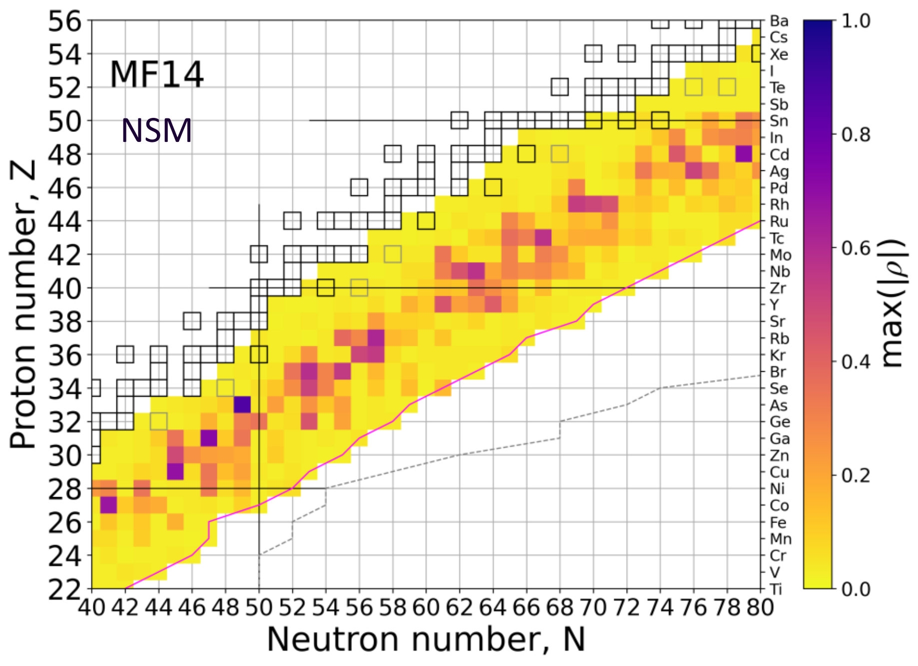
Elemental abundance — isotopic neutron capture rate correlation

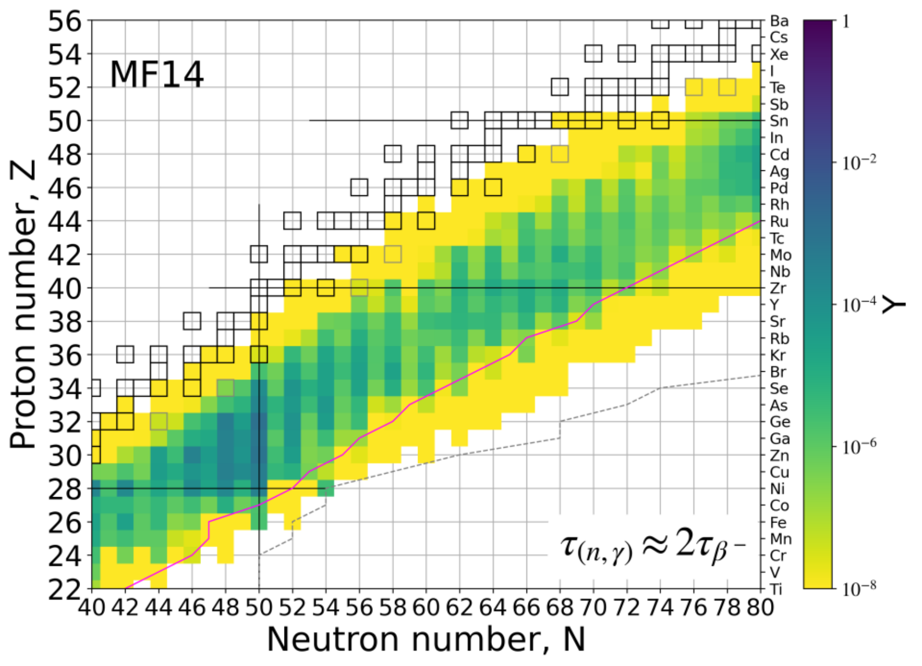




^{89}Se impacts Zr due to detailed balance: $\lambda_{\gamma,n}^{Z,A+1} \propto \lambda_{n,\gamma}^{Z,A}$

Most influential neutron capture rates





Some of the neutron capture rates on isotopes that are populated when neutron capture and beta decay are competing are most influential to elemental abundances.

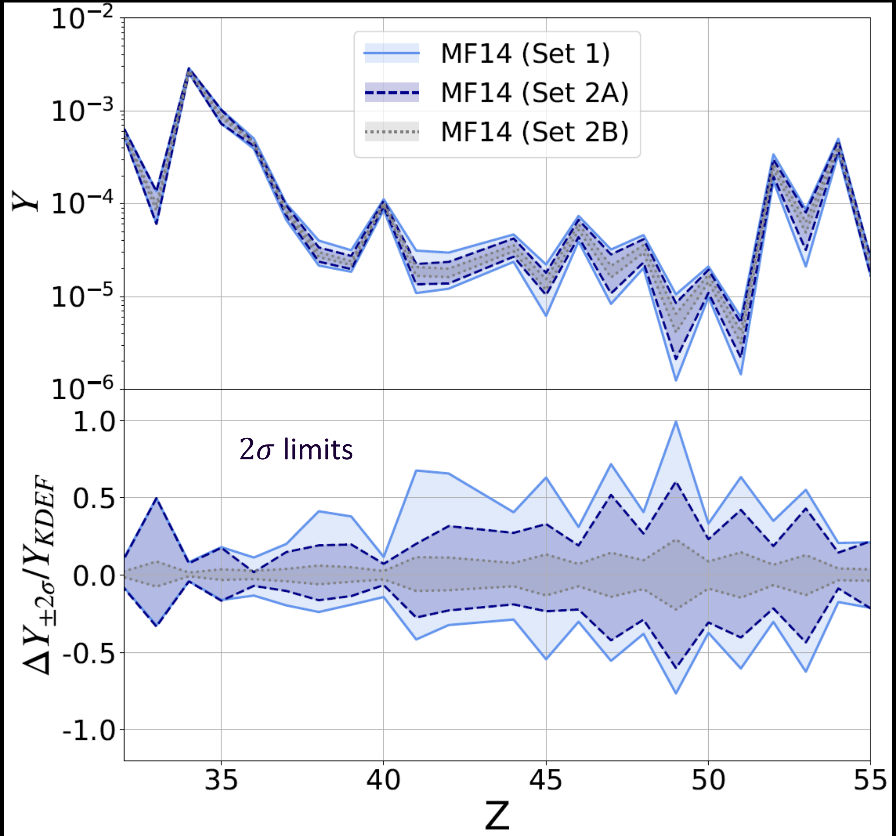
Reduction in neutron capture rate uncertainties

Set 1   : Default (n,γ) uncertainties

Set 2A : Reduce (n,γ) uncertainties on rates for only **35** isotopes

Set 2B: Reduce (n,γ) uncertainties on rates for all **1000**

35 correlated isotopes lead to more reduction in uncertainty than the other 1000 do.



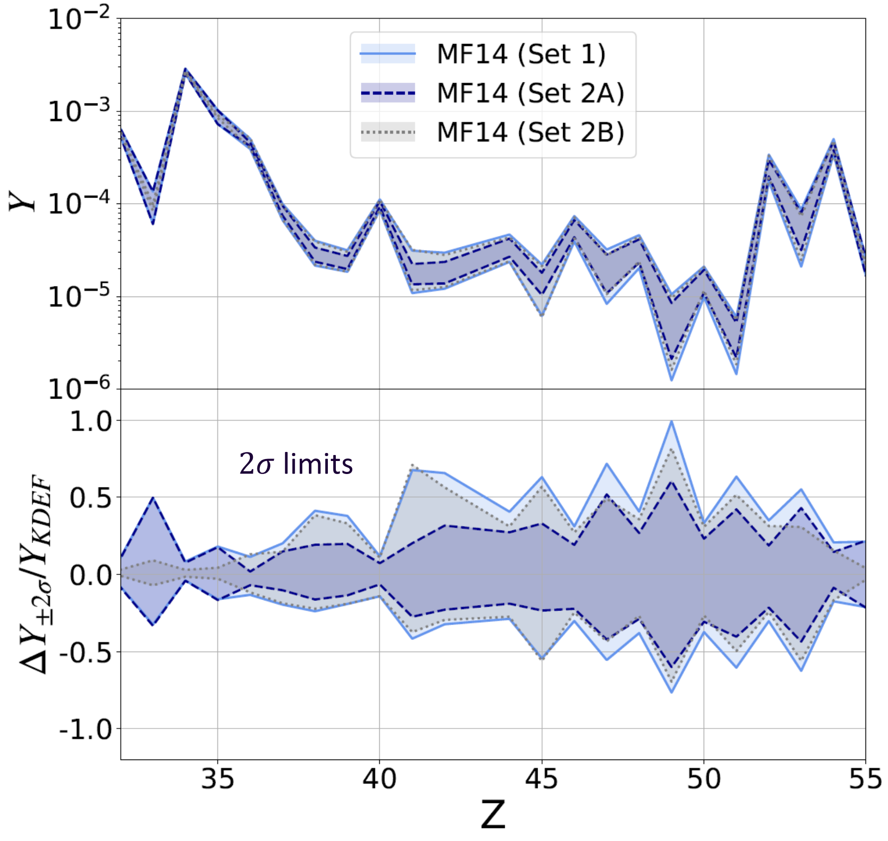
Reduction in uncertainties

Set 1 : No reduction

Set 2A: Reduce uncertainties on rates for only **50** isotopes

Set 2B: Reduce uncertainties on rates for all **1000** but not for the 50 isotopes

35 correlated isotopes lead to more reduction in uncertainty than the remaining 1000 do



Astrophysical sites for the *r* process

Binary Neutron star mergers:
Meyer+1989, Metzger & Fernandez 2014, Just+2015, Radice+ 2018, Lund+2024

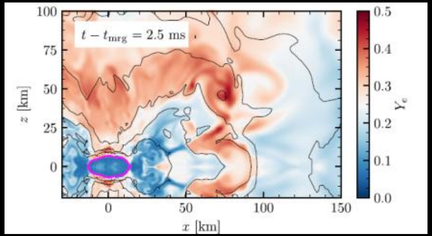
Neutron star-black hole mergers:
Surman+2008, Darbha+2021, Curtis+2023

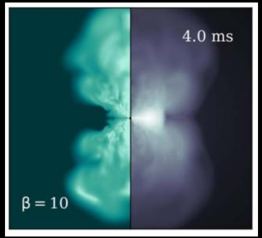
Magnetorotational supernovae:
Nishimura+2015, Mosta+2018, Reichert+2021, Zha+2024

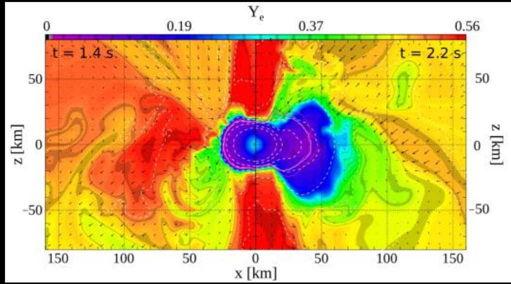
Magnetar giant flares:
Patel+2025a, Patel+2025b

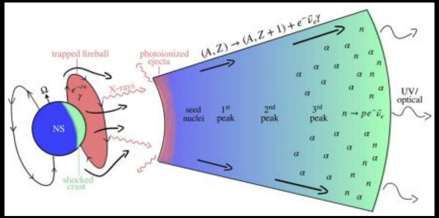
Collapsars:
Siegel+2019, Anand+2024

Accretion-induced or Merger-induced white dwarf collapse:
Pitik+2026a, 2026b.

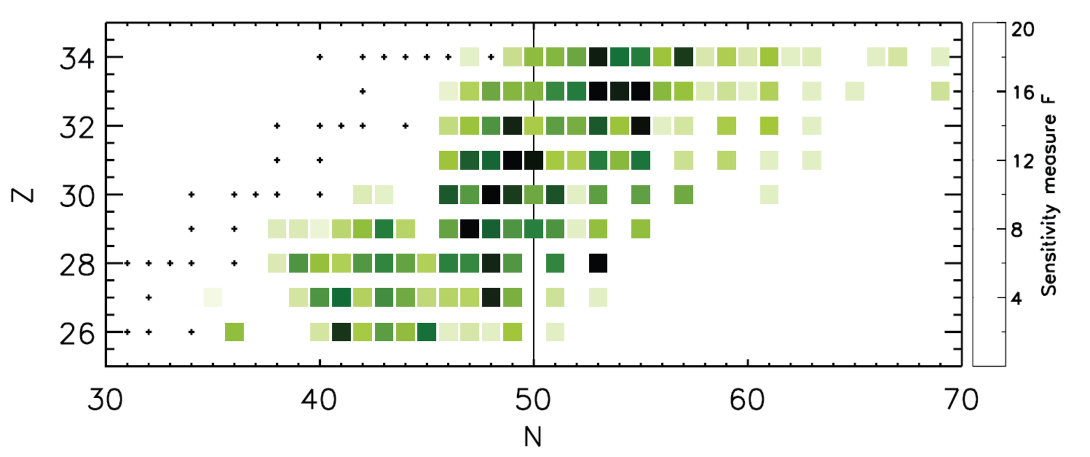








F-metric studies for the full weak *r* abundance yield



Surman *et al*, AIP Adv. (2014), Vescovi et al. FASS (2022)